# بيان Bayan AI

A clinical AI system that interprets between doctors and patients in Qatar, reasons about the consultation, and flags what gets missed.

Team QH-129 Qai × Hugging Face Hackathon O1 Doha, 5–7 Nov 2026

بيان Bayan

# Meet Anurag

29, a Nepali construction worker in Doha

## “My chest is burning. It’s just *gyastrik*.”

ग्यास्ट्रिक, Nepali for “gastric”

The doctor hears reflux.

**It could be his heart.**



# Language barriers in Qatar

**65%**

of HMC ambulance staff face language barriers at work

**71%**

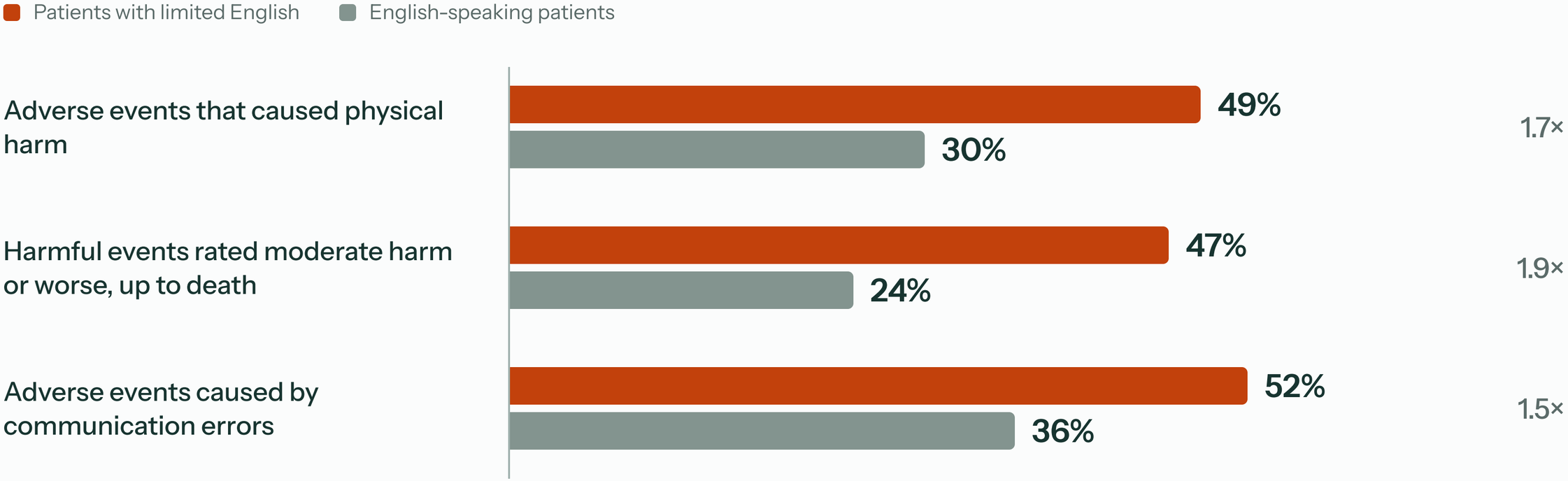
are comfortable letting family members interpret

**9**

languages with face-to-face interpreters at Sidra; Nepali, Malayalam, Bengali and Tagalog are not among them

# Adverse events and language

Many patients in Qatar don’t share a language with their doctor. US hospitals that measured the effect found this.




Bayan is built to close this gap: it interprets, reasons about the consultation and flags what gets missed.

# What Bayan does

It listens to both sides of the consultation.

## Interprets

37 patient languages. The doctor speaks English or Arabic.

## Reasons

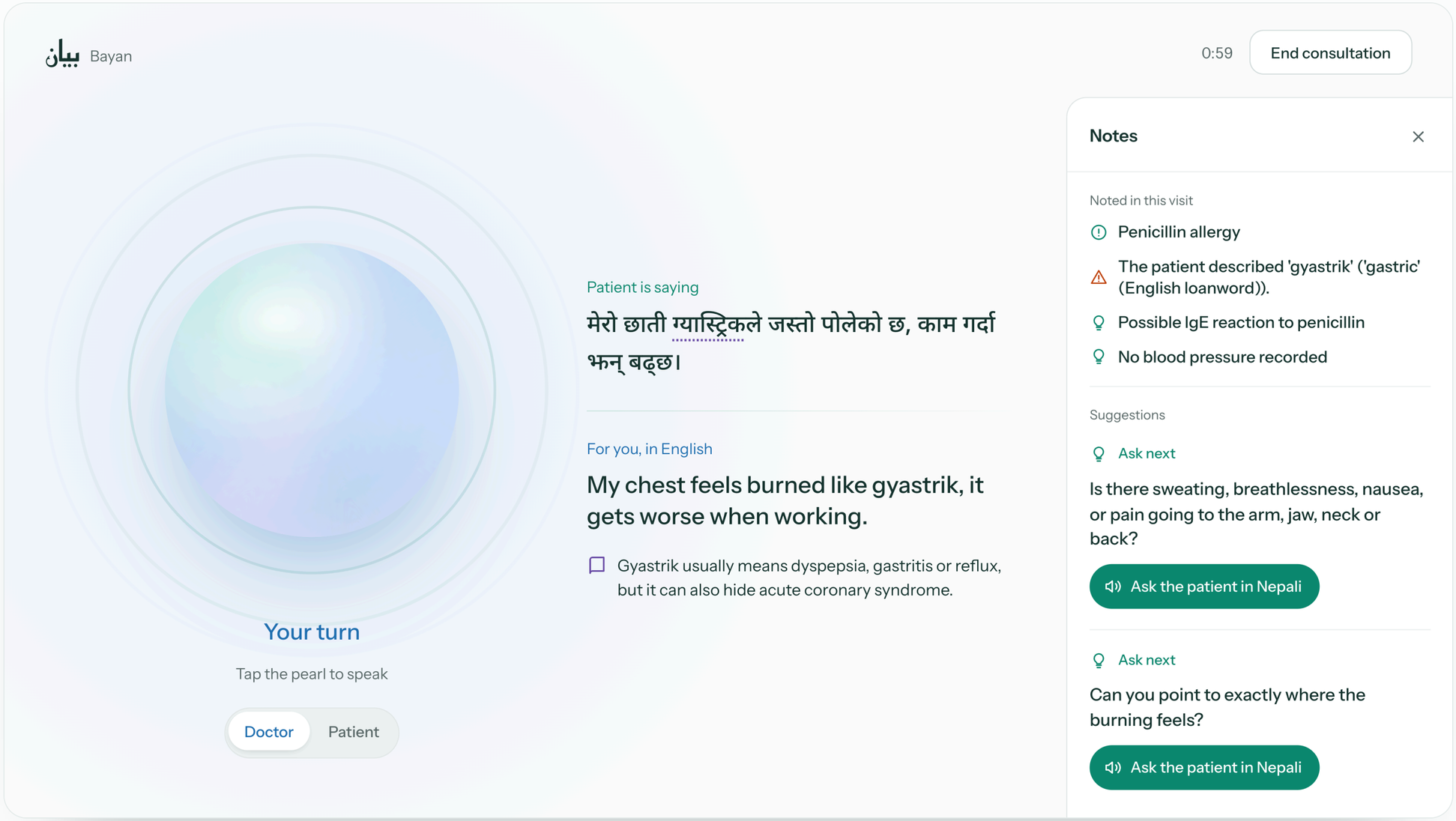
Keeps a running case summary and reviews it after every turn.

## Flags

Allergies, doses, interactions, missed workup, plans that don’t fit the complaint, patient confusion, mood cues.

## Suggests

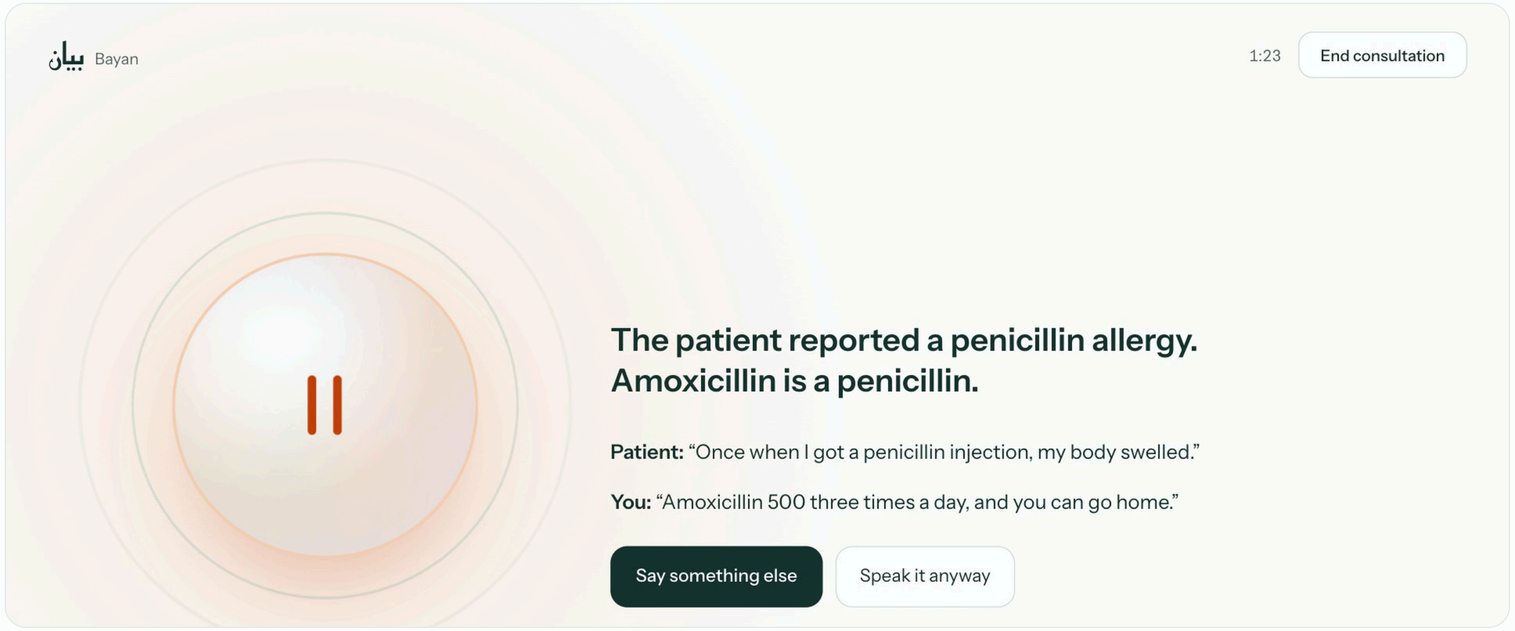
One next question at a time, asked in the patient’s language with one tap.




## Doctor and patient take turns tapping the pearl. Bayan translates, speaks, and checks.

Recorded on the live Space: scripted Nepali consultation, live translation and voices. Played at 1.5× speed.

# Check before it is said




The patient mentioned a penicillin allergy, and the doctor then prescribed amoxicillin. Bayan paused the sentence before the patient heard it, so the doctor can say something else or speak it anyway.

# Care plan and doctor’s note



بيان Bayan

## तपाईंको औषधि र हेरचाह निर्देशन

Your medicines and care instructions

07 Nov 2026

PDF डाउनलोड गर्नुहोस्
Download PDF

**तपाईंको औषधि** Your medicines

**प्यारासिटामोल 500 मि.ग्रा. चक्की**
Paracetamol 500 mg tablet
5 दिनसम्म खाना खाएपछि दिनको 3 पटक 2 चक्की खानुहोस्।
Take 2 tablets 3 times a day after food, for 5 days.

**ओमेप्राजोल 20 मि.ग्रा. क्याप्सुल**
Omeprazole 20 mg capsule
14 दिनसम्म हरेक बिहान खाना अघि 1 क्याप्सुल।
1 capsule every morning before breakfast, for 14 days.

**दैनिक तालिका** Daily schedule

| | |
|---|---|
| **उठ्ने बित्तिकै**<br>When you wake up | ओमेप्राजोल · 1 क्याप्सुल |
| **बिहानको खाना**<br>Breakfast | प्यारासिटामोल · 2 चक्की |

Phone page, via QR code



## உங்கள் மருந்துகள் மற்றும் பராமரிப்பு வழிமுறைகள்

Your medicines and care instructions

07 Nov 2026

### உங்கள் மருந்துகள்
Your medicines

**அசித்ரோமைசின் 500 மி.கி.** Azithromycin 500 mg
500 மி.கி. ஒரு முறை தினம் எடுத்துக்கொள்ளவும். 3 நாட்களுக்கு.
Take 500 mg once a day. For 3 days.

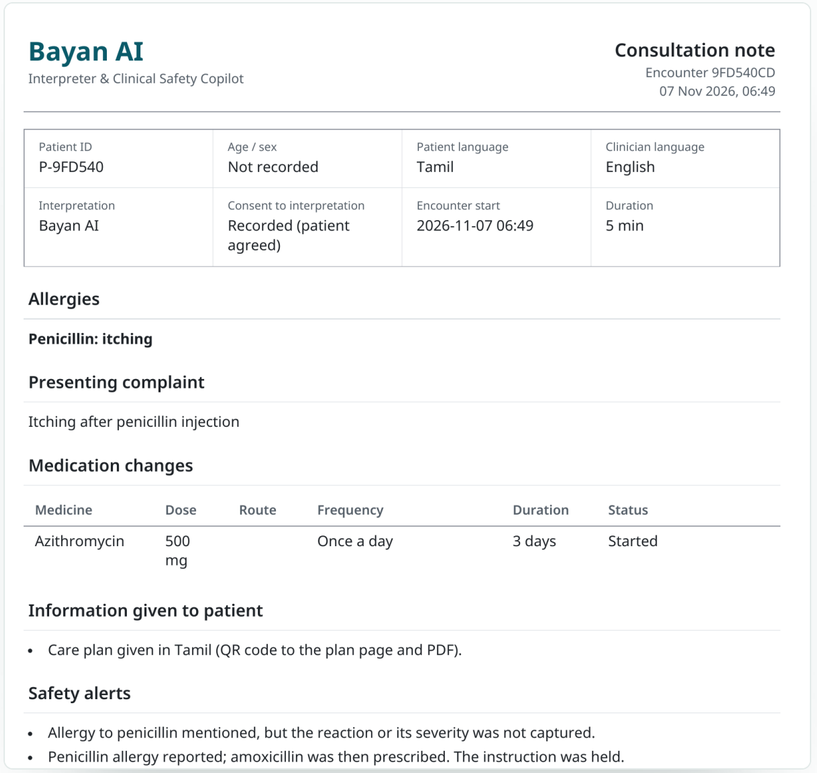
### தினசரி அட்டவணை
Daily schedule

| மருந்து<br>Medicine | காலை உணவு<br>Breakfast |
|---|---|
| **அசித்ரோமைசின்**<br>Azithromycin | **500 மி.கி.**<br>500 mg |

### எடுக்க வேண்டாம்
Do not take

- பெனிசிலின் மருந்துகள், உதாரணமாக அமோக்சிசிலின். நீங்கள்

Patient sheet in Tamil: penicillins under “Do not take”



**Bayan AI**
Interpreter & Clinical Safety Copilot

**Consultation note**
Encounter 9FD540CD
07 Nov 2026, 06:49

| Patient ID | Age / sex | Patient language | Clinician language |
|---|---|---|---|
| P-9FD540 | Not recorded | Tamil | English |
| **Interpretation** | **Consent to interpretation** | **Encounter start** | **Duration** |
| Bayan AI | Recorded (patient agreed) | 2026-11-07 06:49 | 5 min |

### Allergies
**Penicillin: itching**

### Presenting complaint
Itching after penicillin injection

### Medication changes

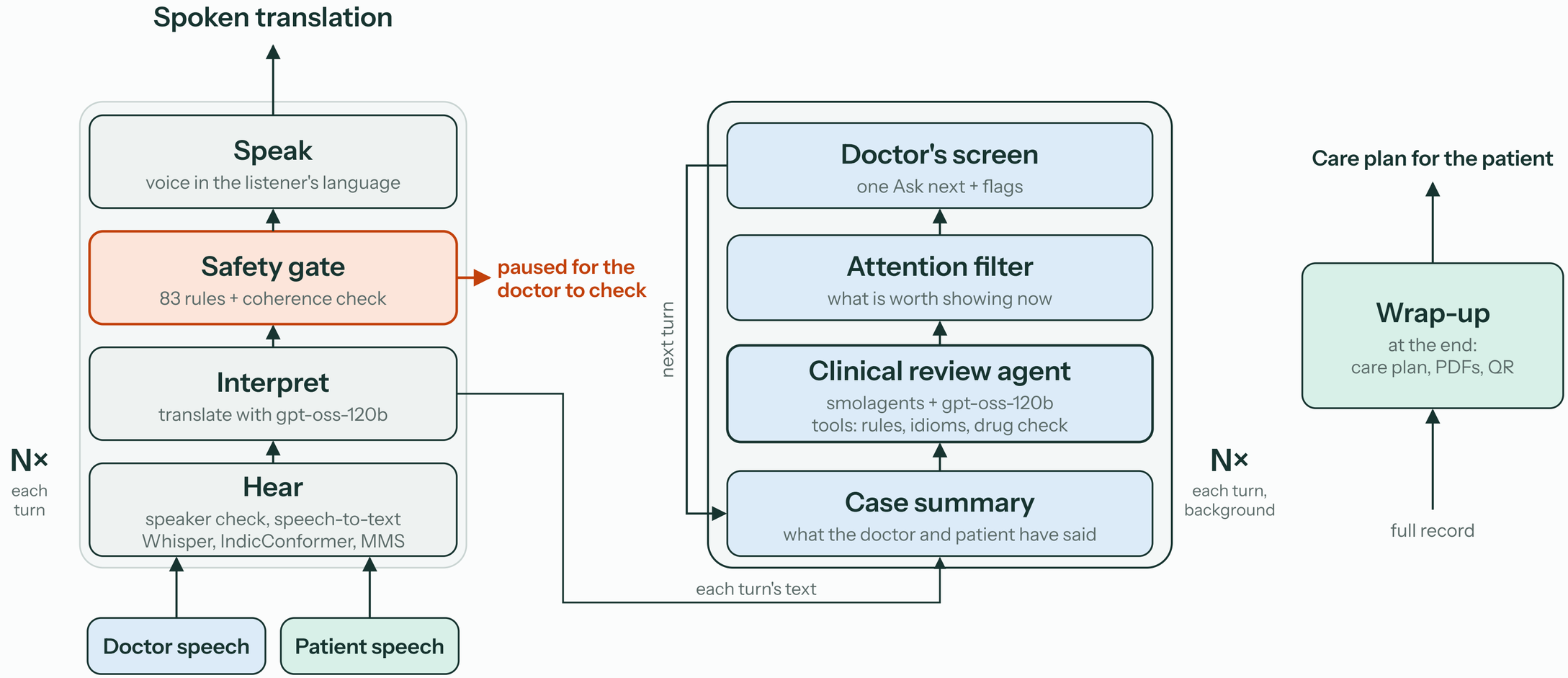
| Medicine | Dose | Route | Frequency | Duration | Status |
|---|---|---|---|---|---|
| Azithromycin | 500 mg | | Once a day | 3 days | Started |

### Information given to patient
- Care plan given in Tamil (QR code to the plan page and PDF).

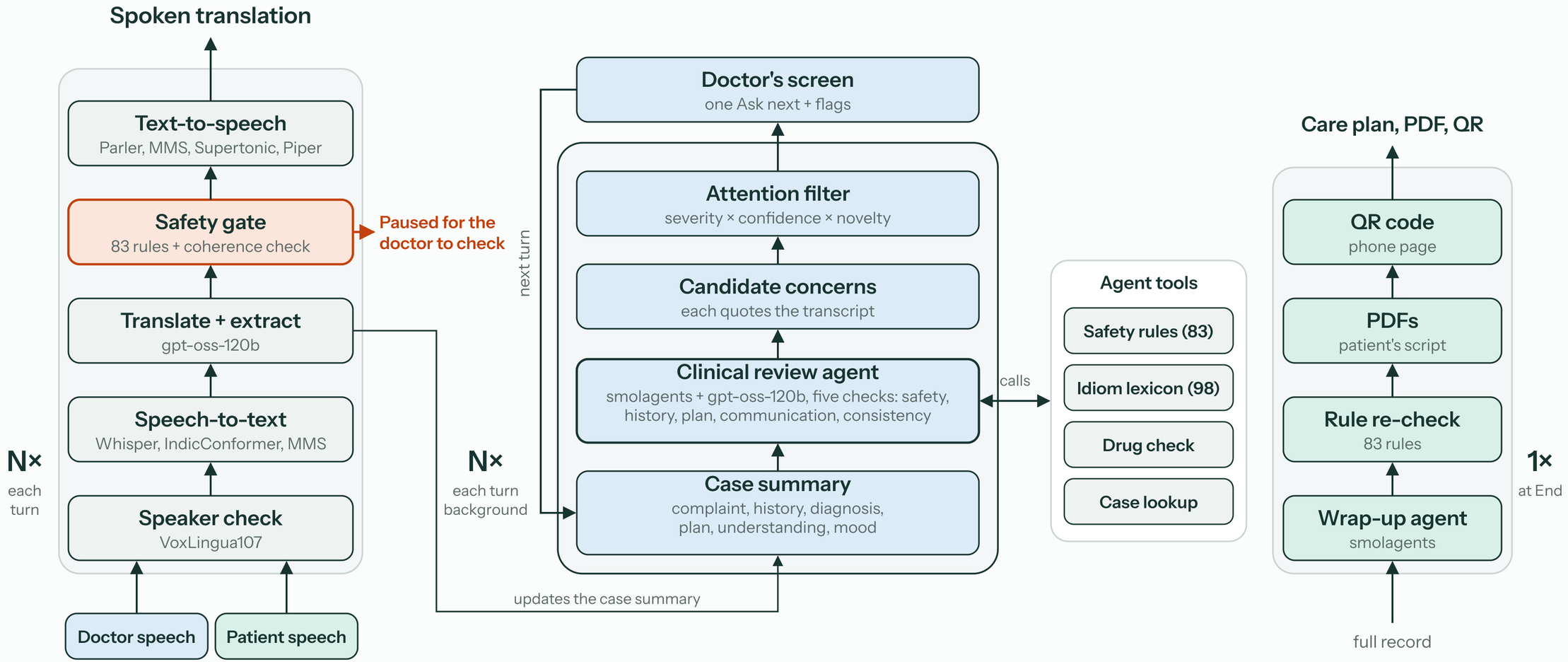
### Safety alerts
- Allergy to penicillin mentioned, but the reaction or its severity was not captured.
- Penicillin allergy reported; amoxicillin was then prescribed. The instruction was held.

Doctor’s note: the paused amoxicillin, on the record

# Architecture (overview)



# Architecture



# Hugging Face stack

## Bayan AI

CPU upgrade

dullgrey/team-qh-129-bayan-app

- gradio-app **gradio**
- huggingface **smolagents**

## Bayan Voice Worker

NVIDIA L4

dullgrey/team-qh-129-bayan-voice

- openai **whisper-large-v3-turbo**
- facebook **mms-1b-all**
- ai4bharat **indic-parler-tts**
- Supertone **supertonic-3**
- ai4bharat **indic-conformer-600m-multilingual**
- speechbrain **lang-id-voxlingua107-ecapa**
- facebook **mms-tts-tgl, mms-tts-swh**
- rhasspy **piper-voices**

## Inference Providers

Cerebras, Groq

router.huggingface.co

- openai **gpt-oss-120b**

# Simulation results

Scripted consultations run through the real engine as text, with gpt-oss-120b. Results vary between runs.

## Safety rules

38 consultations, 14 languages

**99%** of 727 checks passed

**0 of 44** false-alarm traps tripped

**100%** of 477 PDF checks: native script, paused drug never on the patient sheet

## Clinical review

16 consultations, 9 languages: 7 with a plan that doesn't fit, 6 quiet visits, 3 repeats of rule cases

**8 of 12** expected concerns flagged

**0** false caution alerts on the 6 quiet visits

**30 of 30** rule checks still passed, 0 of 22 traps tripped

Translation and gate: 0.9 s per turn. Agent card: about 7 s after the turn. LLM cost: about $0.014 per consultation.

# First customers in Qatar

**1.46 M**

visits to the four QRCS workers' health centres in 2025

**1.56 M**

HMC emergency visits in 2025, adult and paediatric

**5.17 M**

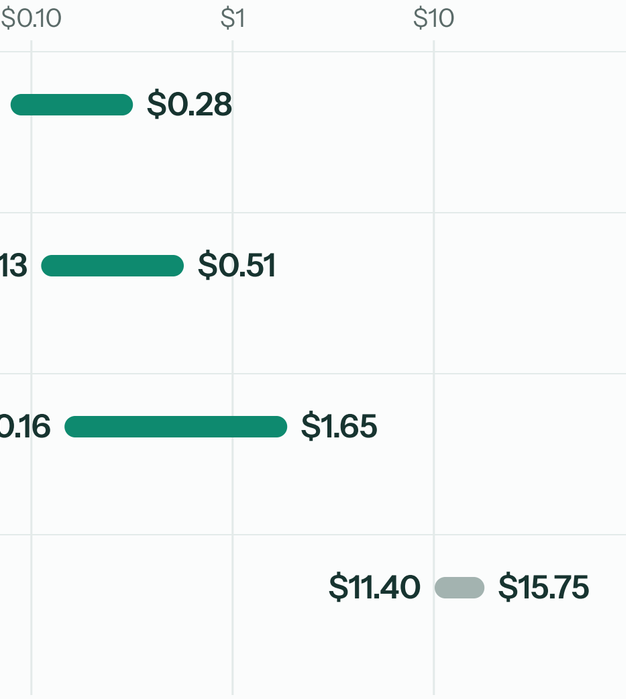
visits to PHCC's 31 health centres in 2024

Pricing to test: QAR 2–3 per consultation. A phone interpreter costs about QAR 40–60 per visit.

# Deployment and cost

All models are open-weight, so the system can also run on a hospital's own servers.

| Where it runs | Up-front hardware | Per clinic-month | Per consultation, log scale |
|---|---|---|---|
| **Online on Hugging Face**<br>Voice on one NVIDIA L4 Space ($0.80/h), app on CPU ($0.03/h), gpt-oss-120b per token on Cerebras | None | $107 if 4 clinics share<br>$339 alone | **$0.09** – **$0.28** |
| **Clinic box**<br>2× NVIDIA L4, about $2,800 each: all voice models plus gpt-oss-20b, on site | $14.6k<br>QAR 53k | $607 alone<br>$152 each if 4 share | **$0.13** – **$0.51** |
| **Hospital, in-country**<br>1× H100 80 GB, about $27.5k, runs gpt-oss-120b (MXFP4, one GPU); 1× L40S, about $8.5k, for voice | $48k<br>QAR 175k | $1,980 per server<br>$198 per clinic across 10 | **$0.16** – **$1.65** |
| **Phone interpreter**<br>CyraCom or Propio $0.76/min, LanguageLine $1.05/min; 15 minutes a visit | None | $13,680–18,900 | **$11.40** – **$15.75** |



# Comparison

| | Nepali, Malayalam, Bengali in the room | Checks allergy, dose, missed workup | Pauses a sentence for the doctor to check | Can run in-country | Cost per visit |
|---|---|---|---|---|---|
| **Bayan** | **Yes, 37 languages** | **Yes, 83 cited rules** | **Yes** | **Yes, open weights** | **$0.09–0.28** |
| Phone and video interpreters | Yes, remote | Human judgement | No | No | $11–74 |
| Google, Microsoft Translate | Text; speech unverified | No | No | No | Free |
| AI medical interpreters | Claimed 40–150+ | Translation checks only | No | Some (Mabel) | Not public |
| Ambient AI scribes | No, English notes | No, documentation | No | No | $150–600 per doctor-month |

# Next steps

A pilot at one QRCS workers' health centre

Clinicians to review our 83 rules and 98 idioms

Native speakers to validate each language

**Team QH-129**
huggingface.co/spaces/dullgrey/team-qh-129-bayan-app

# Team

Team QH-129

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GC

**George Chkhaidze**

# Thank you

Questions

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